# C1. XOR Convenience (Easy Version)

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: bitmasks, constructive algorithms, math | https://codeforces.com/problemset/problem/2189/C1

**This is the *easy* version of the problem. The difference between the versions is that in this version, $2 \le i \le n-1$. Note that a correct solution for the *hard* version is not necessarily a correct solution for the *easy* version.**

Given a natural number $n$. Find a permutation[□] $p$ of length $n$, such that for each $i$ (style*color:red*$2 \le i \le n$-1) there exists a $j$ (style*color:red*$i \le j \le n$) such that $p_i = p_j \oplus i$ [†].

It can be proven that under the constraints of the problem, there exists at least one suitable permutation $p$.

[□]A permutation of length $n$ is an array consisting of $n$ distinct integers from 1 to $n$ in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array), and [1,3,4] is also not a permutation ($n$=3 but there is 4 in the array).

[†]$\oplus$ denotes the bitwise XOR operation.

## Input

Each test contains multiple test cases. The first line contains the number of test cases $t$ ($1 \le t \le 10^4$). The description of the test cases follows.

The only line of each test case contains a single integer $n$ ($3 \le n \le 2 \cdot 10^5$) — the length of the permutation.

It is guaranteed that the sum of $n$ across all test cases does not exceed $2 \cdot 10^5$.

## Output

For each test case, output $n$ integers $p_1, p_2, \ldots, p_n$ — the permutation $p$.

If there are multiple solutions, you may output any of them.

## Example

| Input |
| --- |
| 2<br>3<br>6 |

| Output |
| --- |
| 2 1 3<br>3 6 2 5 1 4 |

## Note

In the first test case, the permutation $p$ = [2,1,3] is suitable since $p_2 = 1$ and $p_3 \oplus 2 = 1$.

In the second test case, the permutation $p$ = [3,6,2,5,1,4] is suitable, as:

- $p_2 = 6 = 4 \oplus 2 = p_6 \oplus 2$

- $p_3 = 2 = 1 \oplus 3 = p_5 \oplus 3$
- $p_4 = 5 = 1 \oplus 4 = p_5 \oplus 4$
- $p_5 = 1 = 4 \oplus 5 = p_6 \oplus 5$